NDEJJE UNIVERSITY

Fear of God brings Knowledge and Wisdom

**COURSE: COMPUTER SCIENCE**

**SEMESTER: II**

**COURSE UNIT: Cryptography and Cyber Security**

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**Assignment: Network and Communication Security**

**1. Explain the concept of network security and discuss its importance in protecting organizational information systems.**

**Network security** refers to the policies, technologies, and procedures used to protect computer networks and data from unauthorized access, misuse, modification, or destruction.

**Importance**

• Protection of sensitive information such as financial records, employee data, and customer information.

• Prevention of cyber-attacks such as hacking, malware, and unauthorized access.

• Maintaining business continuity by preventing service disruption.

• Compliance with security regulations and standards like ISO 27001 and NIST.

• Protecting organizational reputation and maintaining customer trust.

**2. Describe common network communication security threats, including Eavesdropping, Man-in-the-middle attacks, Denial of Service (DoS) attacks, and Malware-based network attacks.**

Eavesdropping

Eavesdropping occurs when an attacker secretly listens to network communications in order to capture sensitive information. This often happens on unsecured networks where data is transmitted without encryption.

Example: A hacker connected to a public Wi-Fi network intercepts login credentials sent over an unencrypted website.

Man-in-the-Middle (MITM) Attacks

In this attack, a malicious actor intercepts communication between two parties and may alter the data being exchanged.

Example: A hacker creates a fake Wi-Fi hotspot. When users connect, the attacker captures usernames, passwords, and credit card details.

Denial of Service (DoS) Attacks

A DoS attack floods a network or server with excessive requests, causing it to slow down or crash so legitimate users cannot access it.

Example: Attackers send thousands of requests to a website server until it becomes unavailable.

Malware-Based Network Attacks

Malware refers to malicious software designed to damage or exploit computer systems. Types include viruses, worms, ransomware, and spyware.

Example: A network worm spreads automatically across multiple computers and disrupts organizational systems.

**3. Discuss the role of network security standards and frameworks such as those recommended by National Institute of Standards and Technology - NIST in securing communication systems.**

The National Institute of Standards and Technology (NIST) provides globally recognized cybersecurity guidelines and best practices that organizations use to secure networks and communication systems.

**The NIST Cybersecurity Framework includes five core functions:**

Identify – understanding cybersecurity risks and network assets.

Protect – implementing safeguards such as encryption and access control.

Detect – identifying security incidents through monitoring systems.

Respond – taking action when an attack occurs.

Recover – restoring systems after a cyber-incident.

**Role in Network Communication Security:**

• Provides standardized security practices for organizations.

• Helps organizations manage and reduce cybersecurity risks.

• Guides implementation of security technologies like firewalls and intrusion detection systems.

• Supports incident response and recovery planning.

Example:

A company following NIST guidelines may install intrusion detection systems, enforce strong authentication policies, and encrypt sensitive network communication.

**4. Explain the following network security mechanisms:**

**Encryption:**

Encryption converts data into a coded format so that unauthorized users cannot read it.

Example: When users access a banking website using HTTPS, their passwords and transactions are encrypted.

**Authentication:**

Authentication verifies the identity of a user or device before allowing access to the network.

Example: Logging into an email account using a password and a one-time verification code.

**Access Control:**

Access control restricts which users can access certain resources in a network.

Example: Only finance department staff can access financial records in an organization.

**Firewalls:**

A firewall is a security device that monitors and filters network traffic based on predefined security rules.

Example: A company firewall blocks unauthorized external access attempts from the internet.

**Intrusion Detection Systems (IDS):**

IDS monitors network traffic to detect suspicious activities or potential security breaches.

Example: If multiple failed login attempts are detected, the IDS alerts the network administrator.

**5.Analyze how secure communication protocols such as:**

- Secure Sockets Layer (SSL)
- Transport Layer Security (TLS) help protect data transmission over the internet.

Secure communication protocols protect information transmitted between devices on the internet.

**Secure Sockets Layer (SSL)**

SSL was one of the first protocols used to secure data communication between web browsers and servers by encrypting the transmitted data.

**Transport Layer Security (TLS)**

TLS is the improved and modern replacement for SSL. It provides stronger encryption and better security mechanisms.

**How They Protect Data**:

• Encryption – prevents attackers from reading transmitted data.

• Authentication – verifies the identity of servers to prevent impersonation.

• Data Integrity – ensures data is not altered during transmission.

**Example:**

When a user visits a website with HTTPS, TLS encrypts the communication between the browser and the web server.

**6. Discuss the security challenges associated with wireless communication networks such as Wi-Fi and mobile networks.**

Unauthorized Access

Attackers may gain access to poorly secured Wi-Fi networks.

Eavesdropping

Attackers may intercept wireless communication using packet sniffing tools.

Rogue Access Points

Hackers can create fake Wi-Fi networks to trick users into connecting.

Weak Encryption

Older wireless security protocols such as WEP are vulnerable to attacks.

**7. Design a basic secure network architecture for an organization, indicating where security controls should be placed.**

Basic Architecture:

Internet

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Firewall

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DMZ (Web Server, Email Server)

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Internal Network (Employees)

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Database Servers

Security Controls Placement:

Firewall – filters incoming and outgoing traffic.

DMZ – isolates public servers from internal systems.

Internal Network – protected by authentication and access control.

Database Servers – protected with encryption and strict permissions.

Network Monitoring – IDS/IPS systems detect suspicious activity.

**8. Critically evaluate the impact of emerging technologies such as cloud computing and Internet of Things (IoT) on network communication security.**

Cloud Computing

Cloud computing allows organizations to store data and run applications on remote servers managed by third-party providers.

Security Challenges:

• Data breaches

• Misconfigured cloud storage

• Shared infrastructure risks

Solutions:

• Strong identity and access management

• Data encryption

• Security monitoring tools

Internet of Things (IoT)

IoT refers to devices connected to the internet such as smart cameras, sensors, and wearable devices.

Security Challenges:

• Weak device security

• Lack of regular updates

• Large number of connected devices increasing attack surface

Example:

Hackers may take control of insecure IoT cameras to spy on users or launch attacks.

Solutions:

- Secure authentication for devices
- Regular firmware updates
- Network segmentation to isolate IoT devices.